Measuring Inaccuracies in School Attendance Zone Data in the Central Virginia Regional Multiple Listing Service

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Introduction

- A **Multiple Listing Service (MLS)** is an online real estate tool for real estate professionals to share information about for sale homes. The vast majority of residential property sales in America use MLS (Gillen, K. C., Schneider, K., & Sturtevant, L., 2023).
- MLS data is entered by real estate listing agents and large real estate companies, and while most MLS's have compliance officers, the **sheer amount of information** included the housing market, along with agents **familiarity biases** makes its **hard to maintain complete accuracy**. (Palm, 1976)
- Each MLS listing is **assigned a school zone,** however, school attendance zones **shift frequently** to account for population shifts, changing educational demands, and updating infrastructure (Holme, J. J., Diem, S., & Welton, A., 2014).
- The **positive relationship** between **school quality** and **home prices** has been well documented in the literature, and because of this, it is important that **school attendance zones are accurate** in the MLS (Collins, C. A., & Kaplan, E. K., 2017). However, very **little academic literature** examines **the reliability of reported attendance zones within MLS.**
- This study examined MLS property data within **eight jurisdictions** of the **Central Virginia Regional MLS (CVR MLS)** Metropolitan Richmond area.

Research Questions

1. How accurate are MLS property listings assigning school zones?
2. Which school zones are the most prone to error? How is this error distributed geographically?
3. What are the potential causes or reasons as to why MLS properties are being misassigned school zones? Are there any relationships between those errors and school quality?

Methods

Received data from VA CVRMLS for every home sold since 2020 (>130k listings).

Gathered current shapefiles for all school levels in all counties.

Found a timeframe where the attendance zones are constant and narrow data to that period.

Narrowed date to August 20, 2025 to present when recent redistricting took effect (n = 12,524 listings).

Conducted analysis for accuracy and binomial test to examine correlation with school quality.

School Inaccuracy Across Eight Counties

Our research centers around **eight localities** in the **Metropolitan Richmond area**, indicated on our map through black boundary lines. Across the study area there are **134 elementary, 40 middle, and 32 high schools.**

Some districts only have **1 school per level** (Charles City County, New Kent and Powhatan for high school, Goochland for middle and high), **resulting in their low error**.

New Kent and **Richmond City** had the **highest % error** among MLS properties sold in the area.

With this, we can visualize that **error varies significantly between counties** and **occurs at high rates in both urban** (Richmond City) **and rural** (New Kent and Powhatan) **jurisdictions**.

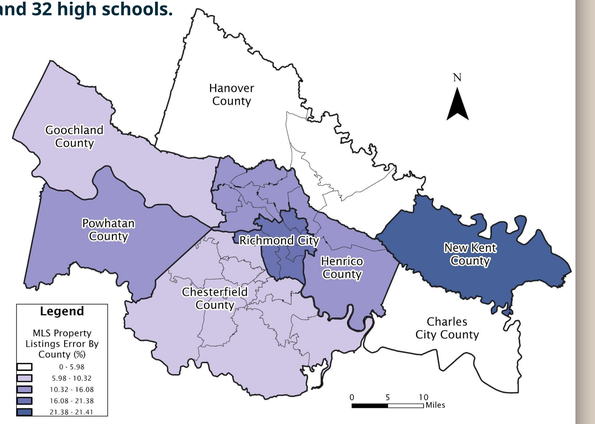


Figure 1. This map shows the distribution of properties as a percentage of the total MLS property listings by county that have been assigned an incorrect school zone by a listing agent. The highest reported error by county was 21.41% in New Kent and 21.38% in Richmond City.

School Inaccuracy by County

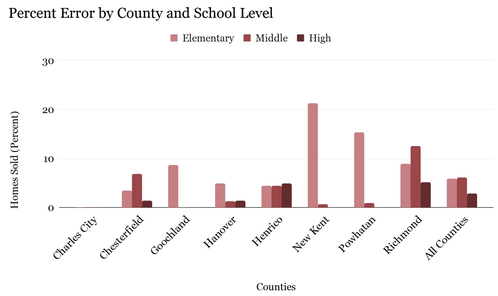


Figure 2. This bar chart demonstrates percent error in each county across each school level, demonstrating highest error in New Kent, Powhatan, and Richmond at the Elementary level.

Powhatan and New Kent counties have the highest rates of inaccuracy likely due to **lack of information or availability**.

- Powhatan **lacks a school assignment locator** and New Kent's locator has a **challenging** user interface.

Changing boundaries also contribute to MLS errors.

- For example, Chesterfield's **recently opened Deep Creek** middle school, and that same school was the **most likely to be misassigned**.

Rural Case Study: Powhatan County

At the **elementary school level**, Powhatan County had the second **highest rate of error** between MLS assignments and actual school zones.

Although only **three elementary schools** exist in the county, there are **large error rates between each** demonstrated through Table 2, suggesting that **real estate agents are not properly aware** or educated on the most recent or **current drawn boundaries**. There is also an issue with a **lack of easy-to-access information** about school attendance zones.

This is made more confusing by the presence of a **Pocahontas Middle School** in [illegible], which some realtors thought Powhatan County had listed as a their assigned middle school.

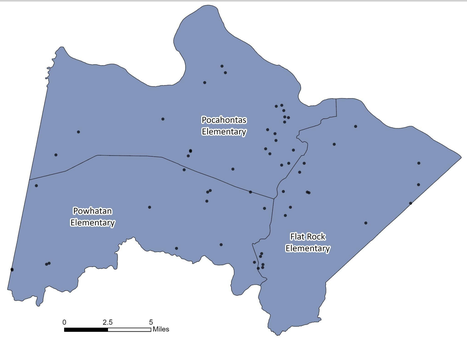


Figure 3. This map shows the distribution of properties in Powhatan County that have incorrectly listed elementary schools. There seems to be a cluster of incorrectly zoned properties closer to the boundary lines between Pocahontas and Flat Rock, however most points show no spatial pattern of distribution.

MLS Assigned	Actual School Zone	N*
Pocahontas	Powhatan	26
Powhatan	Flat Rock	20
Flat Rock	Pocahontas	17

Table 1. This table demonstrates the amount of mismatches occurring across elementary schools in Powhatan.

Urban Case Study: Richmond City

Although **middle schools** across the region had the **highest error rate** compared to elementary and high school zones.

In Richmond City, **12.41% of MLS properties** are **incorrectly** listed to a specific **middle school attendance zone**.

More densely populated neighborhoods show **no spatial pattern of misassignment**, however we can see **properties clustered across school boundary lines** being the **prone to error**.

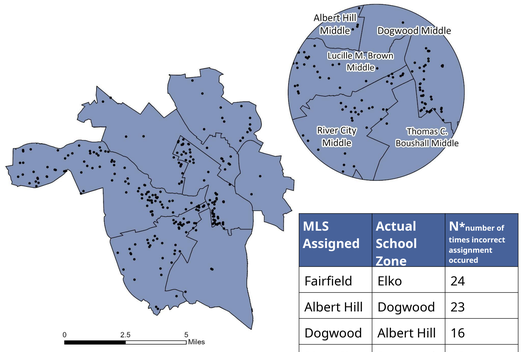


Figure 4. This map shows the distribution of properties in Richmond City that have misassigned middle school zones. It shows a concentration of incorrectly listed properties in pockets close to school boundary lines, primarily between Thomas C. Middle, River City, Lucille M. Brown, Dogwood, and Albert Hill.

MLS Assigned	Actual School Zone	N* number of times incorrect assignment occured
Fairfield	Elko	24
Albert Hill	Dogwood	23
Dogwood	Albert Hill	16
River City	Dogwood	13

Table 2. The accompanying table shows amount of times each specific mismatch occurred. Houses were most often zoned to Elko but assigned to Fairfield.

Inaccuracy Analysis

- The error rate **varies significantly** across counties and **between school levels**. The **highest** error rate was found in **middle schools**, and the **lowest** at **high schools**.
- The **low high school error** is likely because there are **fewer of them**, and therefore it's harder to make a mistake.
- The **high middle school error** rate is likely because middle schools **frequently rezone** and redraw boundary lines to accommodate their less flexible capacity limits
- The **higher elementary school error rate** is mostly likely because there are **more elementaries than any other school**, making the zones smaller and **a mistake easier to make**.
- The **overall error rate of 12.512%** means that about **1 in 8 homes sold** in this data set

	ES	MS	HS	Incorrect Overall Properties
Properties	717	751	348	**1567**/12524
Error Rate %	5.73%	6.00%	2.78%	**12.512%**

Table 3. This table shows the number of errors in MLS school assignment per school level, and the overall error count and rate.

Testing for Quality Bias in MLS School Assignments

- One **hypothesis** for the **cause** of these inaccuracies was the **intentional misrepresentation** of school assignment by real estate agents. Initial research demonstrated **financial incentive** to selling a home in a **higher quality school district** (Carrillo, P., Cellini, S., Green, R., 2013).
- To evaluate this hypothesis, we compared the **quality of MLS-assigned schools** with the **quality of each property's oficial school** using a **one-tailed binomial test**.
  - We used **two publicly available, utilized** measures of school quality; **GreatSchools** (GS) to represent perceived quality, and **Virginia Department of Education School Performance and Support Framework** (SPSF) to represent actual school quality (Feathers, 2020; Matthews, 2025).

School Level	Metric	MLS Higher %	P-Value	Interpretation
Elementary	GreatSchools(GS)	51.25	0.291	No statistical significance
Elementary	School Performance and Support Framework (SPSF)	54.25	0.012	Statistically significant evidence that MLS schools have higher SPSF score
Middle	GreatSchools(GS)	42.91	0.998	No statistical significance
Middle	SPSF	45.94	0.944	No statistical significance
High	GreatSchools(GS)	35.82	1	No statistical significance
High	SPSF	23.53	1	No statistical significance

Table 4. This table demonstrates the difference between MLS-assigned school GS & SPSF scores and the quality of official attendance zone schools. For each school level, binomial tests evaluated whether MLS-assigned schools were more likely than the actual school to have higher GS or SPSF scores than expected by change.

- The results of this test indicate that **there is no statistically significant evidence of systemic intentional misrepresentation** between MLS misassigned schools and actual school assignment.
- The **elementary SPSF scores were significantly higher for MLS assigned schools**, however there was **no observable pattern or distribution of difference**.

Conclusion

Inaccuracies vary greatly across the observed metro-Richmond counties & urban v. suburban areas.

Inaccuracies are most likely caused by human error, boundary complexity, and lack of information.

Most misassignments occur near geographically concentrated school attendance zone boundary lines.

There is no evidence of intentional misrepresentation between school zone quality and misassignment.

Future Work

Future work will examine the **relationship between school quality, time on market, and bidding activity** by incorrectly zoned property. Additional work is also needed to better understand **how MLS reporting practices**, attendance zone changes, and school quality information **shape residential decision making.**

References